

Predicting Patient Length of Stay (PLOS) in Hospitals

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<https://github.com/ErDOS-ProJects/summer26-patient-length-of-stay/tree/main>



Background and project goal



35.6 Million

US ANNUAL ADMISSIONS (AHA)



6 Days

AVERAGE INPATIENT STAY (OECD)

The Double Burden of Prolonged Stay

Clinical Patient Risks

Extended stays increase the acquisition risk of secondary hospital-acquired infections (HAIs), trigger muscle deconditioning, and heighten general psychological distress.

Staff Operational Strain

Prolonged cases limit bed turnover rate, bottlenecking emergency departments and overstressing patient-to-nurse staffing ratios to stressful limits.

Advantages of Predictive Modeling



Dynamic Resource Planning

Foresee upcoming bed vacancies early to schedule nurse shifts and coordinate critical ward staffing schedules optimally.



Enhanced Operational Efficiency

Streamline systemic discharge processes, reduce backlogs, and maximize heavy diagnostic equipment utilization.



Improved Patient Outcomes

Reduce inpatient exposure to high-risk clinical environments, preventing readmissions and refining transitions of care.

Our goal: to develop a robust ML model to predict LoS using only admission-level data, enabling better resource allocation and patient care before a patient's hospital course even begins

Stakeholders and KPI

Our primary stakeholders: patients, hospital systems and administrators, and clinical teams

- Patients – more transparent care timelines and reduced risks of complications tied to prolonged stays
- Hospital administrators – actionable predictions to optimize staffing, bed allocation, and discharge planning
- Clinical teams – reliable estimates to effectively coordinate patient care

Metrics for model evaluation: Root Mean Squared Error, Mean Absolute Error, Huber Loss, and Quantile Loss.

- Five regression models: Linear Regression (baseline), Random Forest, XGBoost, LightGBM and CatBoost
- A model was considered clinically and operationally viable only if it achieved meaningful improvement relative to the baseline

Data source



TEXAS
Health and Human
Services

**Texas Department of State
Health Services**

Texas Inpatient [Public Use Data File \(PUDF\)](#) for 2017-2019 from [Center of Health Statistics](#)

About our data set

- Anonymized and de-identified data on patient discharge from Texas hospitals
- Information on patient demographics, diagnoses, and procedure
- Absolutely massive – about 7 million rows and 167 fields in total

Choosing predictors for model

We choose features/descriptors that can only be obtained at point of admission

Patient-specific

Sex
Race
Ethnicity
Patient age group
Patient zip code
Patient latitude*
Patient longitude*
Patient rural flag*
Distance to provider*

Provider-specific

Public health region
THCIC (provider) ID
Provider latitude*
Provider longitude*
Provider rural status*

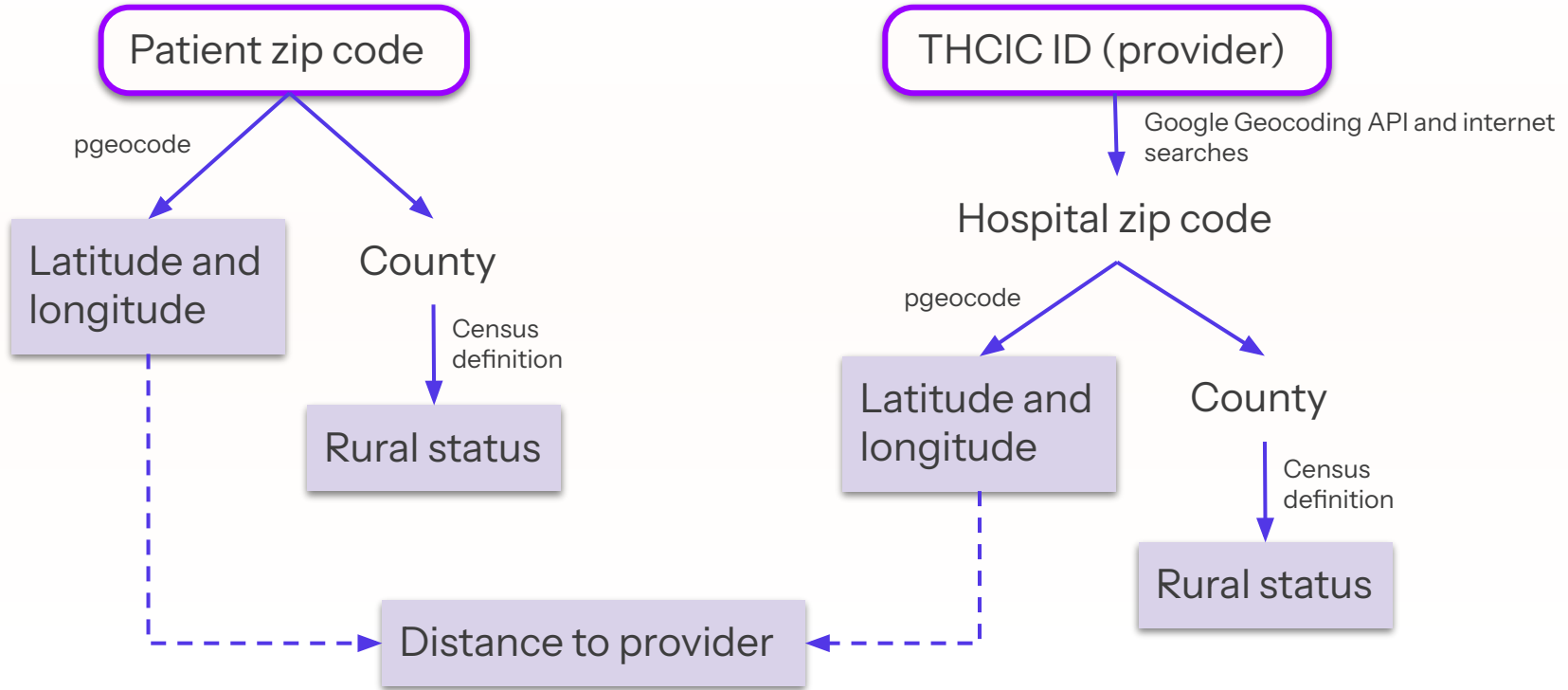
Admission-specific

Type of admission
Source of admission
Weekday of admission
Emergency department flag
Diagnostic codes (chapters 1 to 22)*
Quarter
Year
Number of codes*

* Engineered Data

Target variable: Length of Stay (LoS)

Geographical data processing



Diagnostic code processing

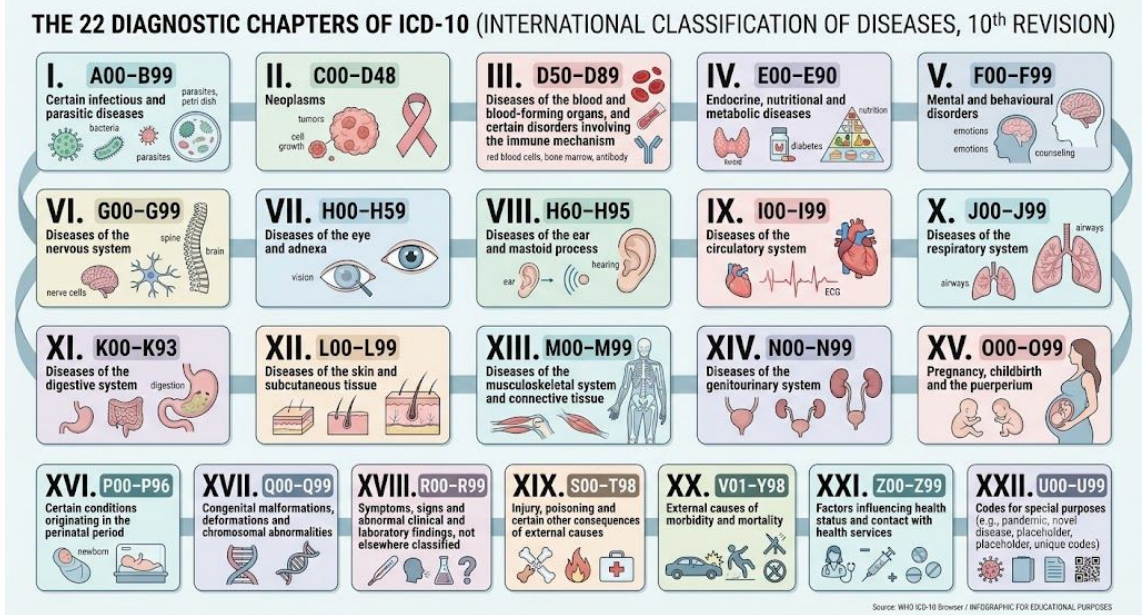
Issue:

There are over 70,000 unique diagnostic codes

A patient can also have multiple diagnostic codes

Simplification:

We convert each diagnostic code into a **diagnostic chapter** of the ICD-10



Data cleaning and processing

Cleaning

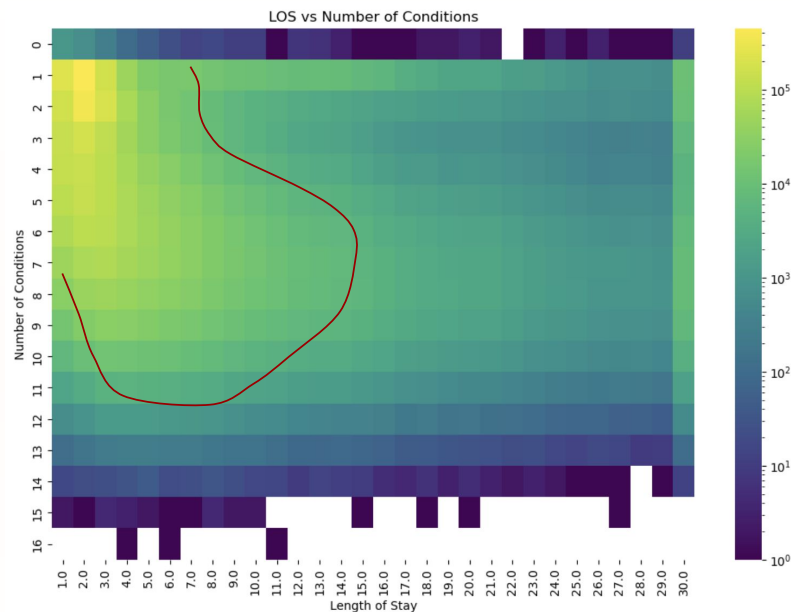
We dropped rows that met one or more of below criteria

- Incomplete or unknown info in at least one data field
- Out-of-state and foreign patients
- Patient-hospital distance is zero

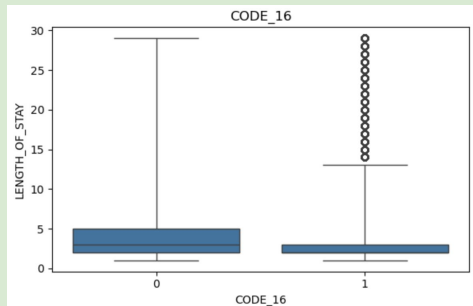
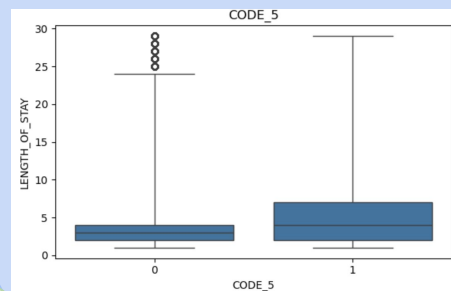
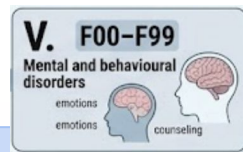
Train-test split

We performed a **80-20 train-test split**, with data stratification by quarter and rural status of patient and hospital

EDA: Diagnostic Codes

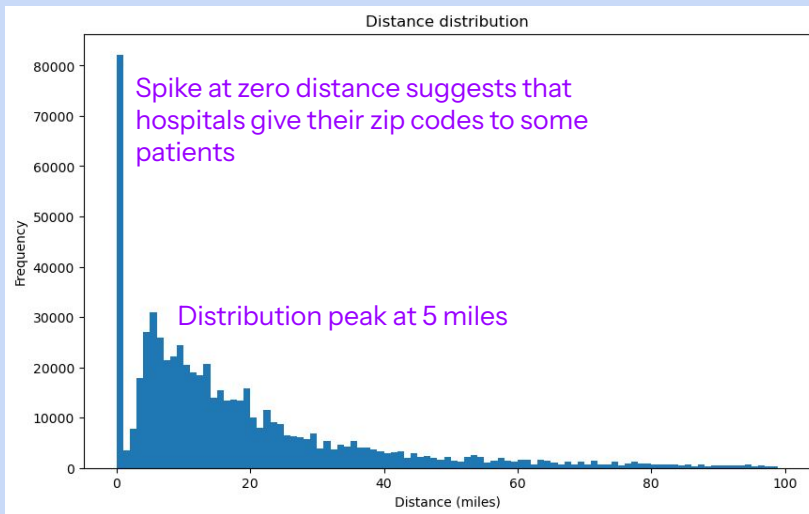


After clipping the length of stay to 30 days, a positive correlation for number of diagnostic codes at admission and length of stay was found, promoting this engineered feature.



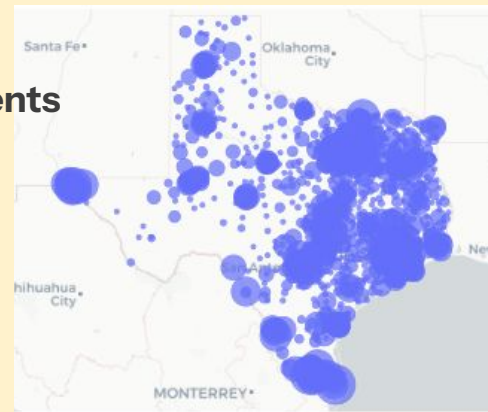
EDA: Geographical distribution

Patients and hospital providers are concentrated in major urban centers (Dallas-Fort Worth, Austin, San Antonio, Houston, Corpus Christi, and El Paso)

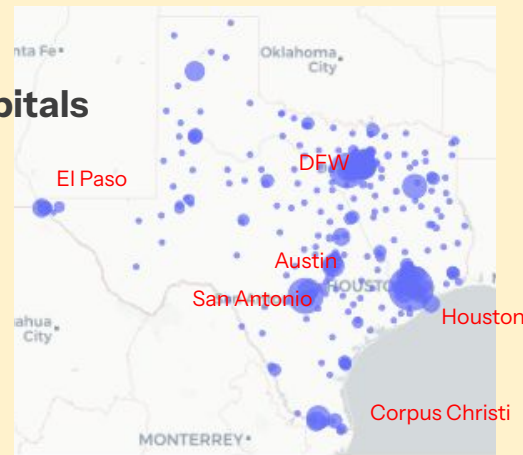


Distribution of patient-to-provider distances

Patients

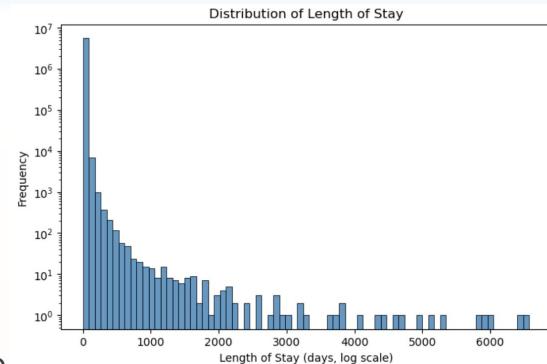


Hospitals



Metrics

Four metrics were chosen based on the large and heavily right-skewed dataset. Each of the four metrics works well with each choice of model.



1) Mean Absolute Error (MAE)

Robust to outliers, easily interpretable, simple and common metric

2) Root Mean Squared Error (RMSE)

Penalizes large errors, easily comparable to MAE, common metric, but not robust to outliers.

(Metric was over-sensitive to outliers in the data set, and models did not perform well)

3) Huber Loss ($\delta = 2.0$)

Combines MAE and RMSE. Errors less than 2 days are handled as RMSE and those greater than 2 days are handled as MAE. Linear Regression showed a MAE of around 4, so $\delta = 2$ was chosen as a halfway point.

4) Quantile ($\alpha = 0.9875$)

Focuses on a certain percentile (chosen to be associated with a 'prolonged' length of stay of 30 days). Heavily penalizes underestimates, representing a conservative metric.

Model Selection

5 models were chosen to compare metrics

1) **Linear Regression (Baseline Model)**

Upside: Simple, fast, and scales well to large datasets.

Downside: Too simple and will miss non-linearity in dataset

2) **Random Forest**

Upside: Simple, robust, requires little tuning

Downside: Less predictive accuracy than boosting methods, potentially slow on large dataset

3) **XGBoost**

Upside: Excellent predictive accuracy

Downside: Requires more tuning and preprocessing

4) **LightGBM**

Upside: Fast and scales well

Downside: Sensitive to tuning and potential to overfit

5) **CatBoost**

Upside: Works very well with many categorical variables

Downside: Slow on large dataset

Each boosting algorithm was trained separately using each of the four optimization metrics, resulting in four distinct models per algorithm.

Chosen Model - CatBoost

CatBoost optimized with RMSE performed the best and outperformed the baseline Linear Regression model in each metric, except RMSE.

Although RMSE is not robust to outliers, all of the boosting models performed best across the 4 metrics when optimized for RMSE. All models struggled with RMSE.

CatBoost runtime was relatively quick when optimized for RMSE (15 minutes on the full training set compared to 1.5 hours for Random Forest)

Model	Optimized w/	RMSE	MAE	Huber	Quantile
Random Forest	RMSE	12.3515	3.2480	4.9660	1.4281
LightGBM		13.8803	2.7116	4.0105	1.8960
XGBoost		13.209	3.0882	4.5028	1.54177
CatBoost	RMSE	13.879	3.063	4.579	1.53076
Linear Regression		12.9841	3.5459	5.4670	1.7715

Percent difference from the Baseline



Model/Metric	Optimized w/	RMSE	MAE	Huber	Quantile
Random Forest	RMSE	-4.87%	-8.40%	-9.16%	-19.38%
LightGBM		6.90%	-23.53%	-26.64%	7.03%
XGBoost		1.73%	-12.91%	-17.64%	-12.97%
CatBoost	RMSE	6.89%	-13.62%	-16.24%	-13.59%

Conclusion

On the Final Testing Set, the model performed similarly to the training set.

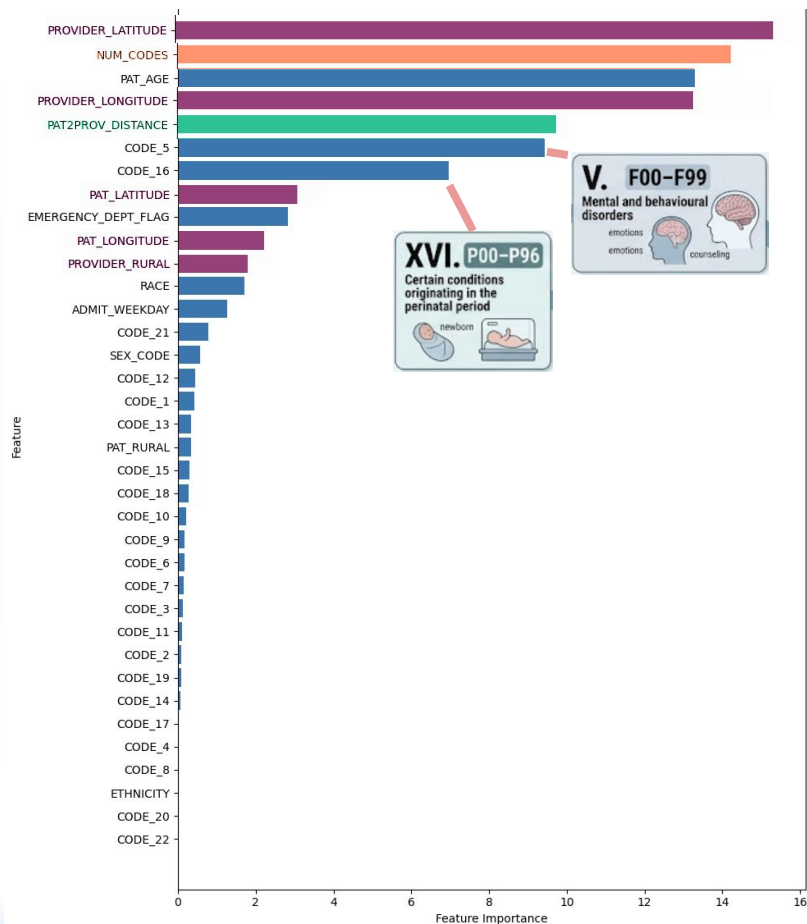
Compared to the Baseline Model, we showed noticeable improvement across all metrics but RMSE (13% to 17% improvement).

6 out of the 10 most important features were engineered

The following features had high relative contribution to the model

- Patient and Provider Location (engineered features)
- Number of Diagnostic Codes at Admission (engineered)
- Distance from Patient to Provider (engineered)
- Diagnostic Code 5 (Mental and Behavioral Disorders) which was found to have a high mean PLoS in EDA.
- Diagnostic Code 16 (Conditions Originating in Prenatal Period) which was found to have a low mean PLoS in EDA.

Feature Importance for CatBoost



Future Steps

- The data was roughly 2% 'prolonged' stays of 30 or more days that were often a challenge to work with. This led to the following ideas
 - Reframe the question as a classification problem for a 'prolonged' stay of 30 or more days.
 - Rework the data set to clip all prolonged stays at 30 days, systematically flattening all outliers. This problem would be focused on trying to predict a 'non-prolonged' stay as accurately as possible (which is the vast majority of the data).
 - A hybrid of the above two ideas: classify the prolonged stays and predict the non prolonged stays exactly
- Analysis could be done on Covid and Post-Covid years to see if there is any meaningful difference in the predictions.
- Subgroup analysis by diagnosis category, patient demographics, and facility type to surface potential disparities in levels of care and identify cases warranting targeted model refinement.

Appendix

EDA: Incidence clustering of codes

Most common diagnostic codes

